

Measure Theory

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1 Measure Theory (測度論)

I σ -algebra or σ -field

i σ -algebra or σ -field

Let X be some set, and let $\mathcal{P}(X)$ represent its power set. A set $\Sigma \subseteq \mathcal{P}(X)$ is called a σ -algebra or σ -field if and only if it satisfies the following three properties:

1. $X \in \Sigma$.
2. Closed under complementation:

$$A \in \Sigma \implies X \setminus A \in \Sigma.$$

3. Closed under countable unions: For all indexed family $(A_i)_{i \in \mathbb{N}} \subseteq \Sigma$,

$$\bigcup_{i \in \mathbb{N}} A_i \in \Sigma.$$

These imply:

$$\emptyset \in \Sigma$$

Proof.

$$X \in \Sigma \implies X \setminus X = \emptyset \in \Sigma.$$

□

These imply closed under finite unions: For all indexed family $(A_i)_{i \in \mathbb{N} \cap [1, n]} \subseteq \Sigma$,

$$\bigcup_{i=1}^n A_i \in \Sigma.$$

Proof. Let $(B_i)_{i \in \mathbb{N}}$ be defined as $B_i = A_i$ for $i \in [1, n]$ and $B_i = \emptyset$ for $i \in (n, \infty)$ and apply closed under countable unions. □

ii Measurable space (可測空間)

A tuple (X, Σ) of a set X and a σ -algebra Σ on X is called a measurable space.

iii Intersection is σ -algebra

Let X be some set, and let $\mathcal{P}(X)$ represent its power set. If $\mathcal{F}$ is a set of σ -algebras of X , then $\bigcap_{\Sigma \in \mathcal{F}} \Sigma$ is a σ -algebra of X .

Proof. If $A \in \bigcap_{\Sigma \in \mathcal{F}} \Sigma$, then $\forall \Sigma \in \mathcal{F} : A \in \Sigma$, then $\forall \Sigma \in \mathcal{F} : X \setminus A \in \Sigma$, then $X \setminus A \in \bigcap_{\Sigma \in \mathcal{F}} \Sigma$.

If $S \subseteq \bigcap_{\Sigma \in \mathcal{F}} \Sigma$ and $|S| \in \mathbb{N} \cup \{\aleph_0\}$, then $\forall \Sigma \in \mathcal{F} : S \subseteq \Sigma$, then $\forall \Sigma \in \mathcal{F} : \bigcup_{A \in S} A \in \Sigma$, then $\bigcup_{A \in S} A \in \bigcap_{\Sigma \in \mathcal{F}} \Sigma$. □

iv σ -algebra generated by collection of subsets

For a set X and a collection $C \subseteq \mathcal{P}(X)$, the σ -algebra generated by C , denoted as $\sigma(C)$ or $\sigma\{C\}$, is the smallest σ -algebra that is a superset of C .

v Borel σ -algebra

Let X be a topological space, the Borel σ -algebra of it, $\mathcal{B}(X)$, is the σ -algebra generated by the set of all Borel sets of X .

vi Borel measurable space

A tuple $(X, \mathcal{B})$ of a set X and a Borel σ -algebra $\mathcal{B}$ on X is called a Borel measurable space.

vii Intervals generating Borel σ -algebra of the real numbers theorem

The Borel σ -algebra of the real numbers can be generated by the collection of all intervals of the form $(-\infty, x]$ for $x \in \mathbb{R}$.

Proof. For every $a < b$, the interval $(a, b]$ can be expressed as

$$(a, b] = (-\infty, b] \cup (-\infty, a]^c.$$

For every $a < b$, the interval (a, b) can be expressed as

$$(a, b) = \bigcup_{n=1}^{\infty} (a, b - 1/n].$$

□

viii π -system or pi-system

A π -system or pi-system on a set Ω is a set P of subsets of Ω such that

- $P \neq \emptyset$, and
- $A, B \in P \implies A \cap B \in P$.

ix Generated π -system

For any non-empty set Σ of subsets of a set Ω , there exists a π -system P , called π -system generated by Σ , that is the unique smallest π -system of Ω containing every element of Σ . It is equal to the intersection of all π -systems of Ω containing every element of Σ .

Proof. TODO

□

x λ -system, lambda-system, Dynkin system, or d-system

A λ -system, lambda-system, Dynkin system, or d-system on a set Ω is a set D of subsets of Ω such that:

- $\Omega \in D$,
- Closed under complements of subsets in supersets: $A, B \in D \wedge A \subseteq B \implies B \setminus A \in D$, and
- Closed under countable increasing union: if $A_1 \subseteq A_2 \subseteq \dots$ is an increasing sequence of sets in D , then $\bigcup_{i=1}^{\infty} A_i \in D$.

xi π -system, λ -system, and σ -algebra

1. A set is both a π -system and a λ -system iff it is a σ -algebra.
2. A π -system is not necessarily a λ -system.
3. A π -system is not necessarily a σ -algebra.
4. A λ -system is not necessarily a π -system.
5. A λ -system is not necessarily a σ -algebra.

Proof. TODO

□

xii (Sierpiński–Dynkin's) π - λ theorem

If P is a π -system and D is a λ -system with $P \subseteq D$, then the σ -algebra generated by P is a subset of D .

Proof. TODO

□

II Measure

i Measure (測度)

Let (X, Σ) be a measurable space. A function $\mu : \Sigma \rightarrow [-\infty, +\infty]$ is called a (positive) measure on Σ iff it satisfies

- Countable additivity, also called σ -additivity: For all indexed family $(E_k)_{k \in \mathbb{N}}$ of pairwise disjoint sets in Σ ,

$$\mu\left(\bigcup_{k \in \mathbb{N}} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

- Nonnegativity:

$$\forall E \in \Sigma : \mu(E) \geq 0,$$

Countable additivity implies $\mu(\emptyset) = 0$.

Proof. By applying for $(\emptyset)_{k \in \mathbb{N}}$,

$$\mu(\emptyset) = \sum_{k=1}^{\infty} \mu(\emptyset)$$

□

Countable additivity implies finite additivity: For all indexed family $(E_k)_{k \in \mathbb{N} \cap [1, n]}$ of pairwise disjoint sets in Σ ,

$$\mu\left(\bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n \mu(E_k).$$

Proof. Let $(F_k)_{k \in \mathbb{N}}$ be defined as $F_k = E_k$ for $k \in [1, n]$ and $F_k = \emptyset$ for $k \in (n, \infty)$ and countable additivity. $\square$

If $\mu(E) \geq 0$ is replaced with $\mu(E) \in \mathbb{R}$, it's called a signed measure.

If $\mu(E) \geq 0$ is replaced with $\mu(E) \in \mathbb{C}$, it's called a complex-valued measure.

ii Measure space (測度空間)

A tuple (X, Σ, μ) of a set X , a σ -algebra Σ on X , and a measure μ (sometimes signed) on Σ is called a measurable space.

iii Pre-measure

Let X be a set and $\mathcal{R}$ be ring of subsets on X . A function $\mu : \mathcal{R} \rightarrow [-\infty, +\infty]$ is called a signed pre-measure on $\mathcal{R}$ iff it satisfies

- $\mu(\emptyset) = 0$ and
- countable additivity, also called σ -additivity, that, for all finite or countable infinite indexed family $(E_k)_{k \in K}$ of pairwise disjoint sets in $\mathcal{R}$,

$$\mu\left(\bigcup_{k \in K} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

If it also satisfies nonnegativity:

$$\forall E \in \mathcal{R} : \mu(E) \geq 0,$$

it is called a pre-measure or positive pre-measure.

iv Outer or exterior measure

Let X be some set, and let $\mathcal{P}(X)$ represent its power set. A function $\mu : \mathcal{P}(X) \rightarrow [-\infty, +\infty]$ is called a signed outer or exterior measure on $\mathcal{R}$ iff it satisfies

- $\mu(\emptyset) = 0$ and
- countable subadditivity, that, for all finite or countable infinite indexed family $(E_k)_{k \in K}$ of pairwise disjoint sets in $\mathcal{P}(X)$,

$$A \subseteq \bigcup_{k \in K} E_k \implies \mu(A) \leq \sum_{k=1}^{\infty} \mu(E_k).$$

If it also satisfies nonnegativity:

$$\forall E \in \mathcal{R} : \mu_0(E) \geq 0,$$

it is called a outer or exterior measure or outer or exterior measure.

v Product measure space

Let $(X, \mathcal{A}, \mu), (Y, \mathcal{B}, \nu)$ be two measure space with possibly signed measure μ, ν . The sigma algebra of the product measure space $X \times Y$ is

$$\sigma(\{A \times B \mid A \in \mathcal{A} \wedge B \in \mathcal{B}\}),$$

denoted as $\mathcal{A} \otimes \mathcal{B}$, and the possibly signed measure of it is $\mu \times \nu$ restricted to $\mathcal{A} \otimes \mathcal{B}$.

vi σ -finite measure

Let X be a set, $\mathcal{R}$ be a ring of sets on X , and μ being a signed or positive pre-measure. μ is σ -finite iff there exists an index family $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{R}$ such that

$$\forall n \in \mathbb{N}: \mu(A_n) < \infty \wedge \bigcup_{n \in \mathbb{N}} A_n = X.$$

If μ is a σ -finite measure, the measure space (X, Σ, μ) is called a σ -finite measure space.

vii Measures agree on generating π -system agree theorem

Suppose μ_1, μ_2 are two measures on a measurable space $(X, \mathcal{F})$, and $\mathcal{F}$ is generated by a π -system P on X . If

$$\forall A \in P: \mu_1(A) = \mu_2(A)$$

and

$$\mu_1(X) = \mu_2(X) < \infty,$$

then $\mu_1 = \mu_2$.

Proof. Define $D = \{A \in \mathcal{F}: \mu_1(A) = \mu_2(A)\}$, which is a λ -system since $\mu_1(X) = \mu_2(X) < \infty$. By π - λ theorem, $\sigma(P) \subseteq D$. □

viii Carathéodory's extension theorem

Let X be a set, $\mathcal{R}$ be a ring of sets on X , μ_0 be a pre-measure on $\mathcal{R}$, and $\sigma(\mathcal{R})$ be the σ -algebra generated by $\mathcal{R}$.

The Carathéodory's extension theorem states that there exists a measure μ on $\sigma(\mathcal{R})$ such that μ_0 is an extension of μ .

Moreover, if μ_0 is σ -finite, then the extension μ is unique and also σ -finite.

Proof. TODO □

ix Measurable functions

Let (X, Σ) and (Y, T) be two measurable spaces. A function $f: D \subseteq X \rightarrow Y$ is said to be measurable iff for every $E \in T$, the preimage

$$f^{-1}(E) \in \Sigma.$$

x Borel measurable functions

Let $(X, \mathcal{B})$ be a Borel measurable space and $(Y, \mathcal{T})$ be a measurable spaces. A function $f : D \subseteq X \rightarrow Y$ is said to be Borel measurable iff for every $E \in \mathcal{T}$, the preimage

$$f^{-1}(E) \in \mathcal{B}.$$

xi Null set

A set A in the σ -algebra is called a μ -null set for a measure μ if $\mu(A) = 0$.

xii Almost everywhere (a.e.)

If (X, Σ, μ) is a measure space, a property P is said to hold almost everywhere or μ -almost everywhere in $Y \subseteq X$ if there exists a measurable set $N \in \Sigma \wedge N \subseteq Y$ with $\mu(N) = 0$, such that for all $x \in Y \setminus N$, the property P holds at x .

xiii Continuity from below

Let $(A_n)_{n \in \mathbb{N}}$ be an increasing sequence of measurable sets, i.e., $A_{i-1} \subseteq A_i$ for all i , and let $A = \bigcup_{n=1}^{\infty} A_n$. Then a measure μ satisfies

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Proof. Define

$$B_1 = A_1,$$

$$B_n = A_n \setminus A_{n-1}, \quad n \geq 2.$$

$$\mu(A_n) = \sum_{k=1}^n \mu(B_k),$$

$$\mu(A) = \sum_{k=1}^{\infty} \mu(B_k).$$

□

xiv Continuity from above

Let $(A_n)_{n \in \mathbb{N}}$ be a decreasing sequence of measurable sets, i.e., $A_{i-1} \supseteq A_i$ for all i , and let $A = \bigcap_{n=1}^{\infty} A_n$. Then a measure μ such that $\mu(A_1) < \infty$ satisfies

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Proof. Define

$$B_n = A_1 \setminus A_n.$$

By continuity from below,

$$\mu\left(\bigcup_{k=1}^n B_n\right) = \mu(A_1 \setminus A) = \lim_{n \rightarrow \infty} \mu(A_1 \setminus A_n)$$

Since $A, A_n \subseteq A_1$, by countable additivity,

$$\mu(A_1 \setminus A) = \mu(A_1) - \mu(A)$$

$$\mu(A_1 \setminus A_n) = \mu(A_1) - \mu(A_n)$$

$$\mu(A_1) - \mu(A) = \lim_{n \rightarrow \infty} (\mu(A_1) - \mu(A_n)) = \mu(A_1) - \lim_{n \rightarrow \infty} \mu(A_n)$$

Since $\mu(A_1) < \infty$,

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n)$$

□

III Borel measure

i Borel measure

A Borel measure is a measure on a Borel σ -algebra.

ii Radon measure

Let X be a locally compact Hausdorff space, and μ be a Borel measure of X .

- μ is called inner regular or tight iff, for every open set U , $\mu(U)$ equals the supremum of $\mu(K)$ over all compact subsets K of U .
- μ is called outer regular iff, for every Borel set B , $\mu(B)$ equals the infimum of $\mu(U)$ over all open sets U that contain B .
- μ is called locally finite iff every point of X has a neighborhood U for which $\mu(U)$ is finite.
- μ is called a Radon measure if it is inner regular and locally finite.

IV Lebesgue measure (勒貝格測度)

i Real numbers

Let R be the set of all open intervals, and $\ell((b-a)) = b-a$ be the length function. For any subset $E \subseteq \mathbb{R}$, the Lebesgue outer measure $\lambda(E)$ is defined as

$$\lambda(E) = \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) : (I_k)_{k \in \mathbb{N}} \subseteq R \wedge E \subseteq \bigcup_{k=1}^{\infty} I_k \right\}.$$

ii Real vector space

Let R be the set of all open rectangle in $\mathbb{R}^n$ with $n \in \mathbb{N}$, that is, $\prod_{i=1}^n I_i$ for open intervals I_i . For any $C \in R$, the Lebesgue outer measure $\lambda(C)$ is defined as

$$\lambda(C) = \prod_{i=1}^n \ell(I_i).$$

For any subset $E \subseteq \mathbb{R}^n$ with $n \in \mathbb{N}$, the Lebesgue outer measure $\lambda(E)$ is defined as

$$\lambda(E) = \inf \left\{ \sum_{k=1}^{\infty} \lambda(C_k) : (C_k)_{k \in \mathbb{N}} \subseteq R \wedge E \subseteq \bigcup_{k=1}^{\infty} C_k \right\}.$$

iii Carathéodory criterion

A set $E \subseteq \mathbb{R}^n$ satisfies the Carathéodory criterion iff

$$\forall A \subseteq \mathbb{R}^n : \lambda(A) = \lambda(A \cap E) + \lambda(A \cap (\mathbb{R}^n \setminus E)).$$

A set E that satisfies the Carathéodory criterion is Lebesgue-measurable, with its Lebesgue measure being defined as its Lebesgue outer measure. The set of all such E is the Lebesgue σ -algebra.

$\mathcal{B}(\mathbb{R}^n)$ is a subset of Lebesgue σ -algebra.

A set $E \subseteq \mathbb{R}^n$ that does not satisfy the Carathéodory criterion is not Lebesgue-measurable. ZFC proves that such sets do exist.

Proof. TODO

□

iv Unique Borel measure that is translation invariant

Lebesgue measure is the unique Borel measure that is translation invariant, up to scaling by a positive constant.

Proof. TODO

□

V Hausdorff measure (郝斯多夫测度)

i Hausdorff outer measure

Let (X, d) be a metric space. For any subset $U \subseteq X$, let $\text{diam}(U)$ denote its diameter, that is, $\text{diam}(\emptyset) = 0$ and for all $U \neq \emptyset$,

$$\text{diam}(U) = \sup\{d(x, y) \mid x, y \in U\}.$$

Let S be any subset of X , and δ be a positive real number. Define

$$H_{\delta}^d(S) = \inf \left\{ \sum_{k=1}^{\infty} (\text{diam}(U_k))^d \mid S \subseteq \bigcup_{k=1}^{\infty} U_k \subseteq X \wedge \forall k \in \mathbb{N} : \text{diam}(U_k) < \delta \right\}.$$

Note that $H_\delta^d(S)$ is nonincreasing in δ . Thus, $\lim_{\delta \rightarrow 0} H_\delta^d(S) \in [0, \infty]$. The Hausdorff outer measure $H^d(S)$ is defined as

$$H^d(S) := \lim_{\delta \rightarrow 0} H_\delta^d(S).$$

ii Carathéodory criterion

A set $E \subseteq X$ satisfies the Carathéodory criterion iff

$$\forall A \subseteq X : H^d(A) = H^d(A \cap E) + H^d(A \cap (X \setminus E)).$$

A set E that satisfies the Carathéodory criterion is Hausdorff-measurable, with its Hausdorff measure being defined as its Hausdorff outer measure. The set of all such E is the Hausdorff σ -algebra.

$B(X)$ is a subset of Hausdorff σ -algebra.

Proof. TODO

□

A set $E \subseteq X$ that does not satisfy the Carathéodory criterion is not Hausdorff-measurable.

VI Hahn decomposition theorem

i Hahn decomposition theorem

TODO

ii Jordan decomposition theorem

TODO

VII Radon–Nikodym derivative

i Absolute continuity (AC) of measures

Consider a measurable space (X, Σ) on which two measures μ, ν are defined. If for every ν -measurable set A , $\nu(A) = 0$ implies $\mu(A) = 0$, equivalently $\mu(A) > 0$ implies $\nu(A) > 0$, then we write $\mu \ll \nu$ and say μ is dominated by ν or absolutely continuous w.r.t. ν .

ii Radon–Nikodym theorem and derivative

Consider a measurable space (X, Σ) on which two σ -finite measures μ, ν are defined. If $\nu \ll \mu$, then there exists a Σ -measurable function $X \rightarrow [0, \infty)$ such that for any measurable set $A \in \Sigma$,

$$\nu(A) = \int_A f \, d\mu.$$

If the function f exists, it is uniquely defined up to a μ -null set, that is, if g is another function satisfying the same property, then $f = g$ μ -a.e. The function f is commonly written $\frac{d\nu}{d\mu}$ and is called Radon–Nikodym derivative.

Proof. TODO

□

iii Linearity

If $\nu, \mu \ll \lambda$ and $a, b \in \mathbb{R}_{\geq 0}$, then

$$\frac{d(av + b\mu)}{d\lambda} = a \frac{d\nu}{d\lambda} + b \frac{d\mu}{d\lambda}$$

λ -a.e.

Proof. TODO

□

iv Chain rule

If $\nu \ll \mu \ll \lambda$, then

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda}$$

λ -a.e.

Proof. TODO

□

v Reciprocal rule

If $\mu \ll \nu$ and $\nu \ll \mu$, then

$$\frac{d\mu}{d\nu} = \left(\frac{d\nu}{d\mu}\right)^{-1}$$

ν -a.e.

Proof. TODO

□

vi Integral by substitution

If $\mu \ll \lambda$ and $g \in L^1(X, \mu)$, then

$$\int_X g \, d\mu = \int_X g \frac{d\mu}{d\lambda} \, d\lambda.$$

Proof. TODO

□